Day 45 — System Handover

1. Purpose

This document summarizes the HR Interview Simulation System, its main components, scoring configuration, testing procedure, and operational limitations.

2. Main Components

| Component | Responsibility |

|---|---|

| `interview\_ai/interview.py` | Interview workflow |

| `interview\_ai/follow\_up\_engine.py` | Follow-up question analysis |

| `scoring/hr\_interview\_scoring\_engine.py` | Interview scoring |

| `scoring/hr\_interview\_simulation.py` | Candidate simulation and benchmark comparison |

| `scoring/interview\_summary\_generator.py` | Summary and recommendation generation |

| `data/hr\_interview\_scoring\_configuration.json` | Scoring weights and thresholds |

| `tests/` | Automated regression tests |

| `docs/` | Project documentation |

Refer to the implementation and existing tests before changing module behavior or interfaces.

3. Scoring Configuration

The current configuration specifies:

- Score range: 0–100.

- Answer relevance: 35%.

- Communication: 25%.

- Confidence: 25%.

- Consistency: 15%.

- Excellent threshold: 80.

- Good threshold: 65.

- Moderate threshold: 50.

- Minimum keyword-overlap setting: 0.1.

The scoring weights sum to 100%. Any changes should be reviewed alongside relevant tests and documentation.

4. Environment Setup

Run commands from the repository root in PowerShell.

Activate the existing virtual environment if needed:

```
.\\.venv\\Scripts\\Activate.ps1
```

Install project dependencies when setting up a new environment:

```
python -m pip install -r requirements.txt
```

5. Testing Instructions

Run the full test suite:

```
python -m pytest -q
```

Run focused interview and scoring regression tests:

```
python -m pytest -q tests/test\_day40\_hr\_interview\_simulation.py
tests/test\_day39\_interview\_summary\_generator.py
tests/test\_day37\_hr\_interview\_scoring\_engine.py
```

Review the actual test output and investigate failures before accepting a release.

6. Simulation Entry Point

The `HRInterviewSimulationEngine` in `scoring/hr\_interview\_simulation.py` provides the simulation workflow.

Relevant methods include:

- `simulate\_session()`

- `run\_all\_simulations()`

- `compare\_with\_manual\_evaluation()`

- identify_scoring_inconsistencies()

- generate_improvement_recommendations()

- generate_text_report()

The simulation engine is separate from the repository's basic main.py entry point. Follow the existing tests and implementation when running or extending the demo.

7. Release Checklist

- [ ] Run the full automated test suite.

- [ ] Run focused interview and scoring regression tests.

- [ ] Review generated scores and reports.

- [ ] Confirm documentation matches the implementation.

- [ ] Disclose benchmark limitations.

- [ ] Check for accidental secrets or sensitive candidate information.

- [ ] Review all changes before staging and committing.

- [ ] Confirm the final Git working-tree status.

8. Privacy and Responsible Use

- Use anonymized or synthetic candidate data for demonstrations where practical.

- Minimize the storage of personal information.

- Restrict access to confidential candidate records.

- Retain human recruiter responsibility for hiring decisions.

- Treat AI scores as decision-support information, not definitive judgments.

- Review possible fairness and accessibility risks.

- Do not claim legal compliance without an appropriate documented assessment.

9. Known Limitations

The existing four-profile benchmark reports 25% classification agreement, interview Mean Absolute Error of 16.79, and aptitude Mean Absolute Error of 15.35.

These results do not establish production accuracy. The small predefined benchmark and keyword-based scoring indicators require further evaluation and calibration.

10. Handover Status

This document describes the handover requirements. Final completion should be confirmed only after tests, code review, documentation review, repository checks, and stakeholder acceptance have been completed.